

benchmarking tools ENCODE data correlation for checking sample mate

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15 Aug 2025 at 14:26:31

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1 How to generate correlation input data

```
#Step1: for each sample
#samtools mpileup -f /cfs/klemming/home/q/qunli/qunli/Qun/publicData/reference/hg38/addChr/bwaIndex/hg38
#java -jar /cfs/klemming/home/q/qunli/Software/benchmarkingTools/VarScan.v2.4.6.jar mpileup2snp check/A
#bgzip ATAC.vcf
#bcftools index ATAC.vcf.gz
#bcftools reheader -s <(echo "Sample1_ATAC") ATAC.vcf.gz -o ATAC_renamed.vcf.gz
#bcftools index ATAC_renamed.vcf.gz

#Step2: merge all samples and extract freq
#bcftools merge -Oz --force-samples -o merged_samples.vcf.gz ATAC_renamed.vcf.gz RNA_renamed.vcf.gz snA
#bcftools query -f '%CHROM\t%POS[\t%FREQ]\n' merged_samples.vcf.gz > merged_samples_freq.vcf
```

2 ENV

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure2/")
```

3 libraries

```
library(pheatmap)
```

4 Load data

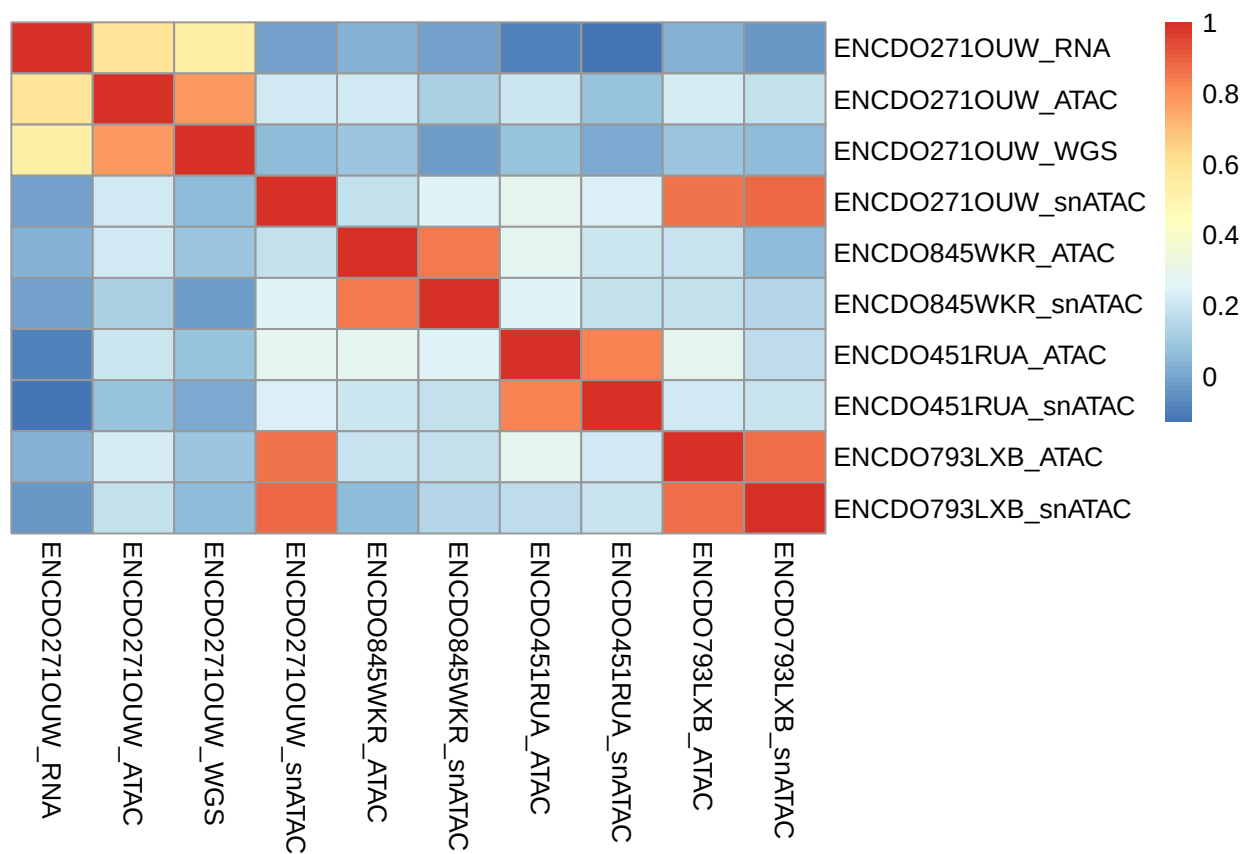
```
VAF <- read.table("../Data/HumanColon/ENCODE_merged_samples_freq.vcf", sep = "\t")
colnames(VAF) <- c("chr", "pos", "ENCDO271OUW_ATAC", "ENCDO271OUW_RNA", "ENCDO271OUW_snATAC", "ENCDO271OUW_WGS",
                  "ENCDO845WKR_ATAC", "ENCDO845WKR_snATAC",
                  "ENCDO451RUA_ATAC", "ENCDO451RUA_snATAC",
                  "ENCDO793LXB_ATAC", "ENCDO793LXB_snATAC")
VAF_frq <- VAF[,c("ENCDO271OUW_RNA", "ENCDO271OUW_ATAC", "ENCDO271OUW_WGS", "ENCDO271OUW_snATAC",
                  "ENCDO845WKR_ATAC", "ENCDO845WKR_snATAC",
                  "ENCDO451RUA_ATAC", "ENCDO451RUA_snATAC",
                  "ENCDO793LXB_ATAC", "ENCDO793LXB_snATAC")]

vaf_num <- VAF_frq
vaf_num <- apply(vaf_num, 2, function(x) {
  x <- gsub("%", "", x)
  x[x=="."] <- NA
  as.numeric(x)
})

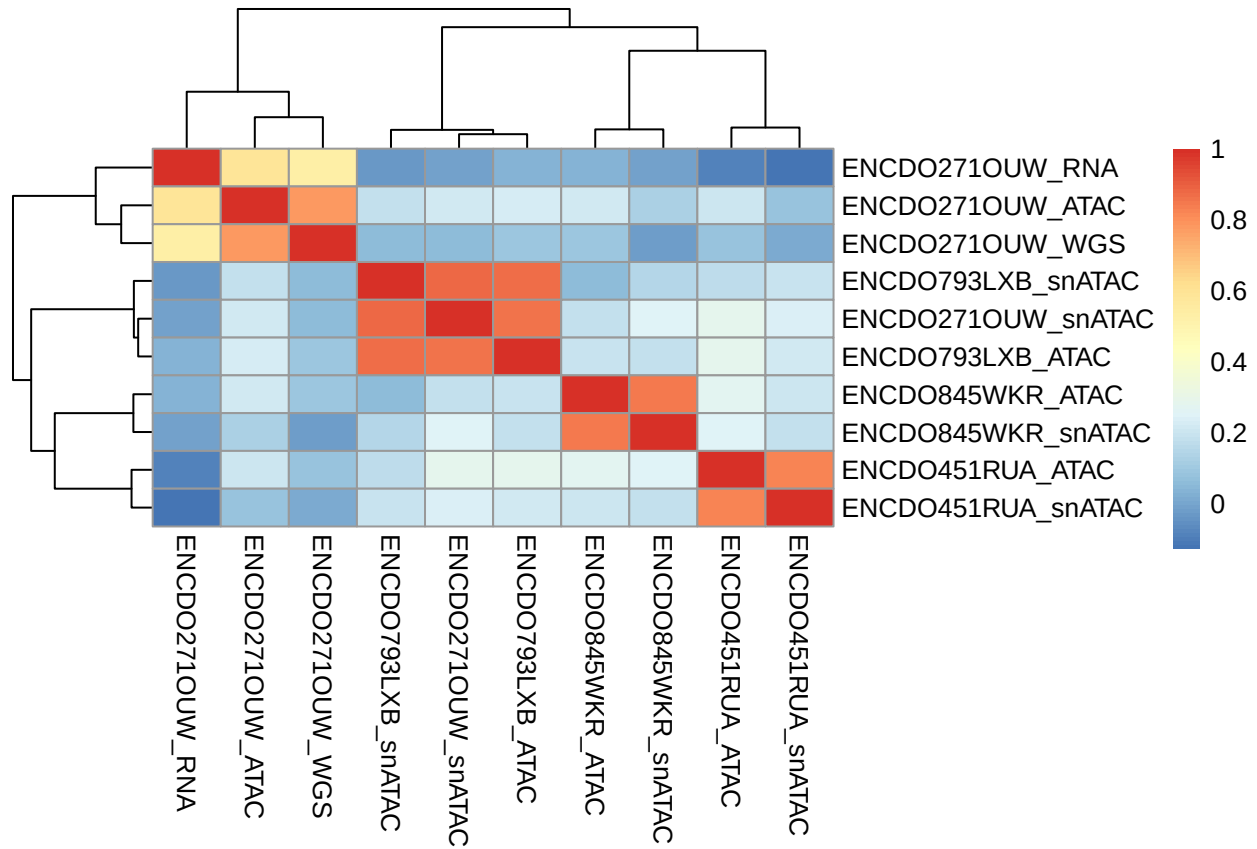
cor_matrix <- cor(vaf_num, use="pairwise.complete.obs", method="pearson")
```

5 plot correlation

```
pheatmap(cor_matrix, cluster_rows = F, cluster_cols = F)
```



```
pheatmap(cor_matrix)
```



SNP-based allele fraction correlation analysis revealed that sample ENCDO271OUW_snATAC showed high genetic concordance with ENCDO793LXB, suggesting a potential sample mislabeling event.